Strong Post-Training from Permissive, Reasoning-Dominant, Web-Scale Pretraining

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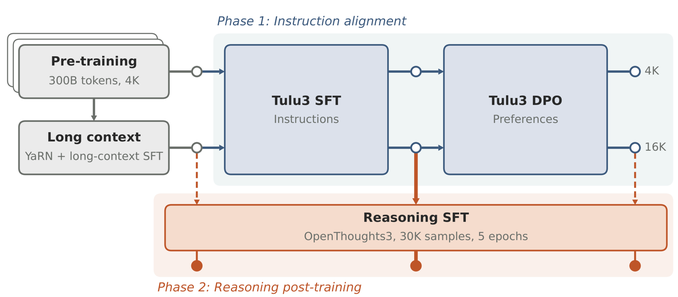
Code, evals & checkpoints

Question & TL;DR

> Can **permissive-first** pretraining keep up with non-permissive web mixtures *after post-training*, if instruction & reasoning data is **front-loaded** into pretraining?

- **Yes, for math, code and reasoning.** Front-loading improves post-trained reasoning for **all four** pretraining datasets, and both enriched permissive mixtures lead the non-permissive ones at 16K.
- **Pretraining decides how much a model gains from post-training.** With front-loading, reasoning SFT gives large gains. Without it, every mixture **declines**.
- **One gap remains:** permissive mixtures still trail on general language understanding.

Controlled setup



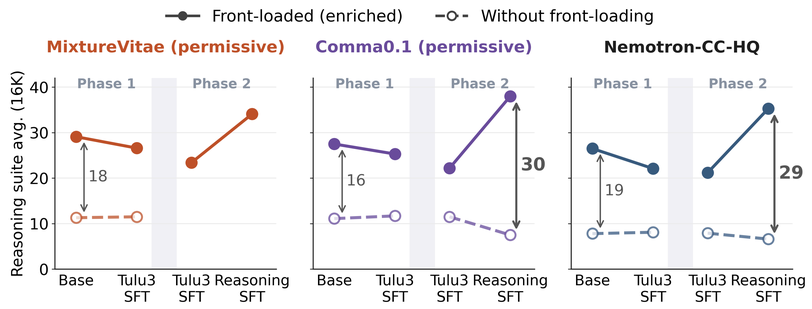
- **1.7B models, 300B tokens**, same recipe, different pretraining data.
- **Evaluation:** 11-task reasoning suite (math, code, science QA, IFEval) + language understanding suite.

1 Front-loading helps every pretraining dataset

> **Adding the instruction & reasoning subset raises the post-trained reasoning suite average in all 24 combinations of dataset × context × stage, by +9.8 to +20.1 points.**

- **Permissive mixtures become competitive.** When every mixture is enriched, both permissive datasets lead both non-permissive ones at 16K after SFT and after DPO, by **3.0–5.1 pts**.
- **Tulu3 cannot substitute for pretraining.** Phase 1 makes front-loaded bases worse on reasoning.
- **Reasoning SFT widens the gap**: front-loaded models gain, models without front-loading decline.
- **One gap persists:** permissive datasets remain **5.5–9.6 pts** behind on the language understanding average.

Front-loading makes models receptive to post-training



2 Post-training: Tulu3 SFT + DPO, then reasoning SFT

> The same 30K-sample reasoning SFT (OpenThoughts3) gives large gains for every **front-loaded** model, and makes every model without front-loading **worse**.

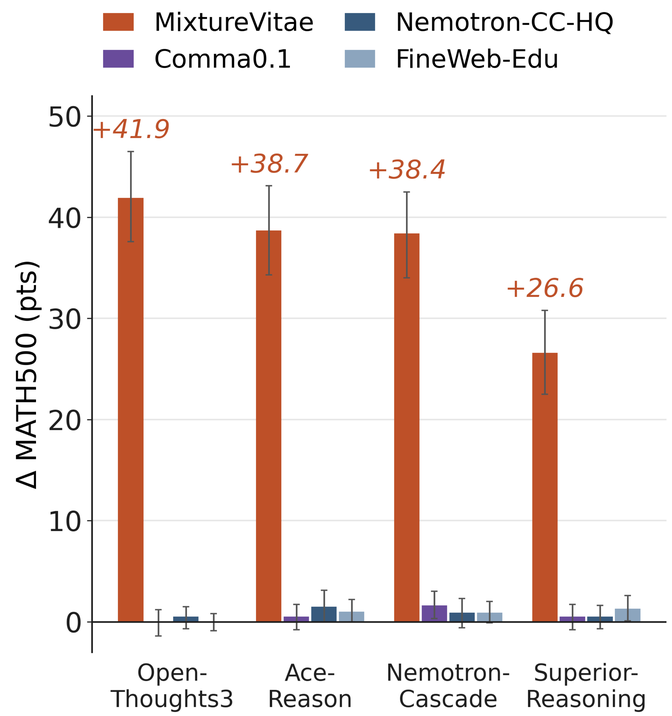
	Phase 1: Tulu3			Phase 2: reasoning SFT		
Reasoning suite avg. (16K)	**Base**	**SFT**	**DPO**	**SFT**	**+ Reas.**	**Δ**
Front-loaded · 300B tokens						
MixtureVitae (perm.)	**29.1**	**26.6**	**26.9**	23.4	34.1	**+10.7**
Comma0.1 (perm.)	27.5	25.3	25.4	22.2	**38.0**	**+15.8**
Nemotron-CC-HQ	26.5	22.1	22.4	21.2	35.3	**+14.2**
FineWeb-Edu	26.0	22.2	21.8	20.0	33.3	**+13.3**
Without front-loading · 300B tokens						
Comma0.1 (perm.)	11.1	11.7	12.0	11.5	7.5	**−4.0**
Nemotron-CC-HQ	7.8	8.1	8.3	7.9	6.6	**−1.3**
FineWeb-Edu	8.3	7.7	8.2	7.5	6.2	**−1.3**
External references · 11–36T tokens						
SmolLM2 1.7B	16.8	18.2	18.7	15.4	11.6	**−3.8**
Qwen2.5 1.5B	9.9	23.4	24.7	23.6	22.9	−0.7
Qwen3 1.7B	30.7	31.5	31.4	28.5	11.3	**−17.2**

Phase 2 is scored with a longer reasoning generation budget (sampled, T = 0.7), so its SFT column differs from Phase 1.

- After Tulu3, **MV** has **more than 2×** the suite average of every matched-compute model without front-loading, and it trained on 37–120× fewer tokens than the external models.
- **MV** goes from 23.4 to **34.1** (MATH500 **71.0**). The enriched **Comma0.1** reaches **38.0**.
- Within each dataset, front-loading adds **+14.6 to +19.8** pts to the reasoning SFT gain.
- Models without front-loading produce very long outputs: 53–100% of their answers come close to the generation limit.
- The language average barely moves (**+0.2** pts for MV).

3 Sensitivity to pretraining data and SFT corpus

The gains do not depend on OpenThoughts3. Three other reasoning datasets also give large MATH500 gains to MV, while the original reference mixtures barely move:



Takeaways

- **Front-loading instruction & reasoning data** makes strong post-training possible under a **permissive data constraint**.
- **Compare pretraining mixtures after post-training**, both their endpoints and how they respond to further training. Base-model rankings can flip.
- **Open problems:** closing the language understanding gap, and keeping instruction following through reasoning SFT.
- **Limitations:** 1.7B scale only, no RL stage, and only a partial overlap audit.

Code, evals & checkpoints: github.com/MixtureVitae/permissive-post-train